

Enterprise Document Intelligence Strategy 2026

Author: Global Data & AI Architecture Team • Classification: Confidential

1. Executive Summary

This strategic document outlines the deployment of unified document intelligence, hierarchical AST chunking, and reciprocal rank fusion search across multi-tenant enterprise repositories. Our objective is to reduce manual contract review times by 65% and achieve sub-50ms hybrid retrieval latencies.

2. Financial Performance & Projection Matrix

The table below details quarterly infrastructure cost optimizations and projected operational return on investment (ROI).

Quarter	Ingestion Volume (Docs)	Avg Latency (s)	Cost Savings (\$)	Accuracy Rate
Q1 2026	250,000	1.42s	\$140,000	98.2%
Q2 2026	480,000	1.18s	\$290,000	99.1%
Q3 2026 (Target)	850,000	0.95s	\$520,000	99.6%
Q4 2026 (Projected)	1,500,000	0.82s	\$950,000	99.8%

3. Architectural Pillars

3.1 Document Parsing with IBM Docling 2.x

By migrating from legacy rule-based OCR to IBM Docling 2.x Unified AST models, DocuFlow preserves complex tabular structures, multi-level headings, bounding boxes, and embedded charts without data loss.

3.2 Dense + Sparse Vector Search Fusion

Semantic similarity embeddings generated via BAAI/bge-small-en-v1.5 FastEmbed models are combined with sparse BM25 lexical rank weights using Reciprocal Rank Fusion (RRF, constant k=60).

4. Enterprise Security & Multi-Tenant Isolation

Every document asset, chunk index, and query context enforces strict cryptographic tenant scoping. Insecure direct object reference (IDOR) vulnerabilities are eliminated through ownership checks at the API, PostgreSQL database, and Qdrant payload collection filter layers.

5. Risk Mitigation & Disaster Recovery

- **Recovery Point Objective (RPO):** Less than 5 minutes via continuous WAL archiving.
- **Recovery Time Objective (RTO):** Less than 15 minutes with automated container failover.
- **Malware Protection:** Streaming magic-byte inspection and EICAR binary signature detection.